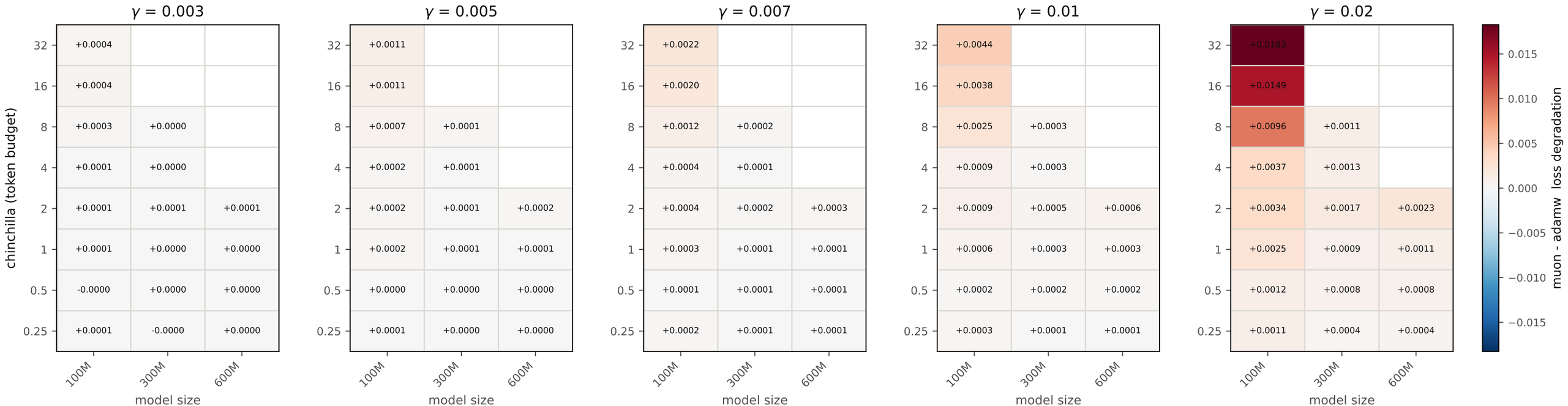
Degradation under Gaussian weight perturbation: muon vs adamw ($\gamma \leq 0.02$), 100M, 300M and 600M
γ = 0.003
γ = 0.005
γ = 0.007
γ = 0.01
γ = 0.02
chinchilla (token budget)
32 16 8 4 2 1 0.5 0.25
+0.0004 +0.0004 +0.0003 +0.0000 +0.0001 +0.0000 +0.0001 +0.0001 +0.0000 +0.0001 +0.0000 +0.0001 -0.0000 +0.0000 +0.0000 +0.0001 -0.0000 +0.0000
+0.0011 +0.0011 +0.0007 +0.0001 +0.0002 +0.0001 +0.0002 +0.0001 +0.0001 +0.0002 +0.0000 +0.0000 +0.0001 +0.0000 +0.0000
+0.0022 +0.0020 +0.0012 +0.0002 +0.0004 +0.0001 +0.0004 +0.0002 +0.0003 +0.0003 +0.0001 +0.0001 +0.0001 +0.0001 +0.0001
+0.0044 +0.0038 +0.0025 +0.0003 +0.0009 +0.0003 +0.0009 +0.0005 +0.0006 +0.0006 +0.0003 +0.0003 +0.0002 +0.0002 +0.0002 +0.0003 +0.0001 +0.0001
+0.0183 +0.0149 +0.0096 +0.0011 +0.0037 +0.0013 +0.0034 +0.0017 +0.0023 +0.0025 +0.0009 +0.0011 +0.0012 +0.0008 +0.0008 +0.0011 +0.0004 +0.0004
100M 300M 600M
model size
muon - adamw loss degradation
0.015 0.010 0.005 0.000 −0.005 −0.010 −0.015
cell = (perturbed - unperturbed) for muon minus the same for adamw | negative (blue) = muon degrades less